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8.19. Theorem. For a compact metric space X , $H^s(X) = \sup \{H^s(C) : C \text{ subset } X \text{ is compact with } H^s(C) < \infty\}$.	Geometric measure theory	<pre> theorem mattila_8_19_compact_subsets_of_finite_hausdorff_measure {X : Type u} [MetricSpace X] [CompactSpace X] [MeasurableSpace X] [BorelSpace X] {s : ℝ} (hs : 0 < s) : muH[s] (Set.univ : Set X) = C : Set X, (_ : IsCompact C), (_ : muH[s] C < inf), muH[s] C := by sorry </pre>
12.14. Theorem. Let A be a Borel set in $\mathbb{R}^n$. (1) If $\dim A > (n+1)/2$, then $L^1(D(A)) > 0$. (2) If $(n-1)/2 < \dim A < (n+1)/2$, then $\dim D(A) > \dim A - (n-1)/2$, where $D(A) = \{x-y : x,y \text{ in } A\}$.	Geometric measure theory	<pre> theorem mattila_12_14_falconer_distance_set {n : ℕ} {A : Set (EuclideanSpace ℝ (Fin n))} (hA : MeasurableSet A) : let D := {r : ℝ exists x in A, exists y in A, r = dist x y} (((n : ℝ) >= 0) + 1) / 2 < dimH A -> 0 < volume D /\ (((n : ℝ) >= 0) - 1) / 2 < dimH A /\ dimH A < ((n : ℝ) >= 0) + 1) / 2 -> dimH A - ((n : ℝ) >= 0) - 1) / 2 < dimH D := by sorry </pre>
6.1. Definition. The upper s -dimensional density of A at x is $\Theta^s(A,x) = \limsup_{r \downarrow 0} (2r)^{-s} H^s(A \cap B(x,r))$. 6.2. Theorem. Suppose $A \subset \mathbb{R}^n$ with $H^s(A) < \infty$. (1) $2^s \Theta^s(A,x) \leq \Theta^s(A,x) \leq 1$ for H^s almost all x in A . (2) If A is H^s measurable, $\Theta^s(A,x) = 0$ for H^s almost all x in $\mathbb{R}^n \setminus A$.	Geometric measure theory	<pre> noncomputable def upperHausdorffDensity {n : ℕ} (s : ℝ) (A : Set (EuclideanSpace ℝ (Fin n))) (x : EuclideanSpace ℝ (Fin n)) : ℝ >= 0 := limsup (fun r : ℝ => muH[s] (A ∩ closedBall x r) / ENNReal.ofReal ((2 * r) ^ s)) (nhds[>] 0) theorem mattila_6_2_hausdorff_density_estimates {n : ℕ} {s : ℝ} {A : Set (EuclideanSpace ℝ (Fin n))} (hs : 0 < s) (hA : muH[s] A < inf) : (forall^m x dmuH[s], x in A -> ENNReal.ofReal (2 ^ (-s)) <= upperHausdorffDensity s A x /\ upperHausdorffDensity s A x <= 1) /\ (MeasurableSet A -> forall^m x dmuH[s], x notin A -> upperHausdorffDensity s A x = 0) := by sorry </pre>
15.3. Definition. A set $E \subset \mathbb{R}^n$ is called m -rectifiable if there exist Lipschitz maps $f_i : \mathbb{R}^m \rightarrow \mathbb{R}^n$, $i = 1, 2, \dots$, such that $H^m(E \setminus \bigcup_i f_i(\mathbb{R}^m)) = 0$. A set $F \subset \mathbb{R}^n$ is called purely m -unrectifiable if $H^m(E \cap F) = 0$ for every m -rectifiable set E . 18.1. Theorem. Let A be an H^m measurable subset of $\mathbb{R}^n$ with $H^m(A) < \infty$. (1) A is m -rectifiable if and only if $H^m(P_V A) > 0$ for $\mathcal{G}(n,m)$ almost all V in $G(n,m)$ whenever B is an H^m measurable subset of A with $H^m(B) > 0$. (2) A is purely m -unrectifiable if and only if $H^m(P_V A) = 0$ for $\mathcal{G}(n,m)$ almost all V in $G(n,m)$.	Geometric measure theory	<pre> def Grassmannian (n m : ℕ) := {V : Submodule ℝ (EuclideanSpace ℝ (Fin n)) // Module.finiteRank V = m} def grassmannianAction {n m : ℕ} (Q : EuclideanSpace ℝ (Fin n) →equiv[ℝ] EuclideanSpace ℝ (Fin n)) (V : Grassmannian n m) : Grassmannian n m := <V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2> def IsInvariantGrassmannianMeasure {n m : ℕ} [MeasurableSpace (Grassmannian n m)] (: Measure (Grassmannian n m)) : Prop := IsProbabilityMeasure /\ forall Q : EuclideanSpace ℝ (Fin n) →equiv[ℝ] EuclideanSpace ℝ (Fin n), (forall x, Q x = x) -> AEMeasurable (grassmannianAction (m := m) Q) /\ Measure.map (grassmannianAction (m := m) Q) = def RectifiableSet (n m : ℕ) (E : Set (EuclideanSpace ℝ (Fin n))) : Prop := exists f : ℕ -> EuclideanSpace ℝ (Fin m) -> EuclideanSpace ℝ (Fin n), (forall j, exists K : NNReal, LipschitzWith K (f j)) /\ muH[(m : ℝ)] (E \ union j, range (f j)) = 0 def PurelyUnrectifiableSet (n m : ℕ) (A : Set (EuclideanSpace ℝ (Fin n))) : Prop := forall E : Set (EuclideanSpace ℝ (Fin n)), RectifiableSet n m E -> muH[(m : ℝ)] (A ∩ E) = 0 theorem mattila_18_1_besicovitch_federer_projection {n m : ℕ} (hm : 0 < m) (hmn : m < n) [MeasurableSpace (Grassmannian n m)] (: Measure (Grassmannian n m)) (h : IsInvariantGrassmannianMeasure) {A : Set (EuclideanSpace ℝ (Fin n))} (hA : MeasurableSet A) (hAfin : muH[(m : ℝ)] A < inf) : (RectifiableSet n m A <-> forall B : Set (EuclideanSpace ℝ (Fin n)), MeasurableSet B -> B subset A -> 0 < muH[(m : ℝ)] B -> forall^m V d, 0 < muH[(m : ℝ)] ((fun x => V.1.orthogonalProjectionOnto x) '' B)) /\ (PurelyUnrectifiableSet n m A <-> </pre>

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		<pre> forall^m V d, muH[(m : R)] ((fun x => V.1.orthogonalProjectionOnto x) '' A) = 0) := by sorry </pre>
7.7. Theorem. Let A subset $\mathbb{R}^n$ and let $f: A \rightarrow \mathbb{R}^m$ be a Lipschitz map. If $m < s < n$, then $\int_A H^s(s-m)(A \cap f^{-1}\{y\}) dL^m y \leq c(n,m) \text{Lip}(f)^m H^s(A)$.	Geometric measure theory	<pre> theorem mattila_7_7_lipschitz_level_sets {n m : ℕ} : exists c : ℝ>=0inf, c < inf /\ forall (s : ℝ) (A : Set (EuclideanSpace ℝ (Fin n))) (f : EuclideanSpace ℝ (Fin n) -> EuclideanSpace ℝ (Fin m)) (K : NNReal), (m : ℝ) < s /\ s < n -> LipschitzOnWith K f A -> (integral^~ y, muH[s - m] (A ∩ f ^-1' {y})) dvolume) <= c * (K : ℝ>=0inf) ^ (m : ℝ) * muH[s] A := by sorry </pre>
Definition. The s-dimensional Hausdorff content of A is $H^s_{\text{inf}}(A) = \inf \sum_i d(E_i)^s$, where the infimum is over all countable covers $A \subset \bigcup_i E_i$. 8.8. Theorem. Let B be a Borel set in $\mathbb{R}^n$. Then $H^s(B) > 0$ if and only if there exists μ in $\mathcal{M}(B)$ such that $\mu(B(x,r)) < r^s$ for x in $\mathbb{R}^n$ and $r > 0$. Moreover, we can find μ so that $\mu(B) > cH^s_{\text{inf}}(B)$, where $c > 0$ depends only on n.	Geometric measure theory	<pre> noncomputable def hausdorffContent {X : Type u} [PseudoMetricSpace X] (s : ℝ) (A : Set X) : ℝ>=0inf := U : ℕ -> Set X, (_ : A subset union i, U i), sum' i, ENNReal.ofReal (diam (U i)) ^ s theorem mattila_8_8_frostman_lemma {n : ℕ} : exists c : ℝ>=0inf, 0 < c /\ c < inf /\ forall (s : ℝ) (B : Set (EuclideanSpace ℝ (Fin n))), 0 < s -> MeasurableSet B -> (0 < muH[s] B <-> exists mu : Measure (EuclideanSpace ℝ (Fin n)), IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu /\ mu != 0 /\ mu B^c = 0 /\ forall x : EuclideanSpace ℝ (Fin n), forall r : ℝ, 0 < r -> mu (closedBall x r) < ENNReal.ofReal (r ^ s)) /\ exists mu : Measure (EuclideanSpace ℝ (Fin n)), IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu /\ mu != 0 /\ mu B^c = 0 /\ (forall x : EuclideanSpace ℝ (Fin n), forall r : ℝ, 0 < r -> mu (closedBall x r) < ENNReal.ofReal (r ^ s)) /\ c * hausdorffContent s B <= mu B := by sorry </pre>
Definition. The Riesz s-energy of μ is $I_s(\mu) = \int \int x-y ^{-s} d\mu(x) d\mu(y)$. The Grassmannian $G(n,m)$ is the space of m-dimensional linear subspaces of $\mathbb{R}^n$, and $\nu_{n,m}$ is its orthogonally invariant probability measure. 9.7. Theorem. Let μ be a Radon measure on $\mathbb{R}^n$ with compact support and with $I_m(\mu) < \infty$. Then $P_{\nu_{n,m}} \mu \ll H^m$ for $\nu_{n,m}$ almost all V in $G(n,m)$ and $\int D(P_{\nu_{n,m}} \mu, u)^2 dH^m u$ $d_{\nu_{n,m}} V < c I_m(\mu)$, where c is a constant depending only on n and m.	Geometric measure theory	<pre> noncomputable def rieszEnergy {n : ℕ} (s : ℝ) (mu : Measure (EuclideanSpace ℝ (Fin n))) : ℝ>=0inf := integral^~ x, (integral^~ y, ENNReal.ofReal (dist x y)^(s-1) ^ s dmu) dmu def Grassmannian (n m : ℕ) := {V : Submodule ℝ (EuclideanSpace ℝ (Fin n)) // Module.finiteRank ℝ V = m} def grassmannianAction {n m : ℕ} (Q : EuclideanSpace ℝ (Fin n) ≃[ℝ] EuclideanSpace ℝ (Fin n)) (V : Grassmannian n m) : Grassmannian n m := <V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2> def IsInvariantGrassmannianMeasure {n m : ℕ} [MeasurableSpace (Grassmannian n m)] (: Measure (Grassmannian n m)) : Prop := IsProbabilityMeasure /\ forall Q : EuclideanSpace ℝ (Fin n) ≃[ℝ] EuclideanSpace ℝ (Fin n), (forall x, Q x = x) -> AEMeasurable (grassmannianAction (m := m) Q) /\ Measure.map (grassmannianAction (m := m) Q) = theorem mattila_9_7_projection_energy {n m : ℕ} [MeasurableSpace (Grassmannian n m)] (: Measure (Grassmannian n m)) (h : IsInvariantGrassmannianMeasure) : exists c : ℝ>=0inf, c < inf /\ forall mu : Measure (EuclideanSpace ℝ (Fin n)), IsFiniteMeasureOnCompacts mu -> Measure.InnerRegular mu -> IsCompact mu.support -> rieszEnergy (m : ℝ) mu < inf -> (forall^m V d, Measure.map (fun x => V.1.orthogonalProjectionOnto x) mu <= </pre>

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		<pre> muH[(m : R)] /\ exists density : forall V : Grassmannian n m, V.1 -> R>=0inf, (forall^m V d, Measure.map (fun x => V.1.orthogonalProjectionOnto x) mu = muH[(m : R)].withDensity (density V)) /\ integral^~ V, integral^~ x, density V x ^ (2 : R) dmuH[(m : R)] d <= c * rieszEnergy (m : R) mu := by sorry </pre>
Definition. The Grassmannian $G(n,k)$ is the space of k-dimensional linear subspaces of $\mathbb{R}^n$, and $\mu_{n,k}$ is its orthogonally invariant probability measure. 10.10. Theorem. Let $m < t < n$ and let A subset $\mathbb{R}^n$ be a Borel set with $0 < H^t(A) < \infty$. Then for all W in $G(n,n-m)$, $H^t(t-m)(A \cap W) < \infty$ for H^m almost all A in W^{bot}, and for $\mu_{n,n-m}$ almost all W in $G(n,n-m)$, $H^m(\{a \text{ in } W^{\text{bot}} : \dim(A \cap W) = t-m\}) > 0$.	Geometric measure theory	<pre> def Grassmannian (n m : N) := {V : Submodule R (EuclideanSpace R (Fin n)) // Module.finiteRank R V = m} def grassmannianAction {n m : N} (Q : EuclideanSpace R (Fin n) equiv[R] EuclideanSpace R (Fin n)) (V : Grassmannian n m) : Grassmannian n m := <V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2> def IsInvariantGrassmannianMeasure {n m : N} [MeasurableSpace (Grassmannian n m)] (: Measure (Grassmannian n m)) : Prop := IsProbabilityMeasure /\ forall Q : EuclideanSpace R (Fin n) equiv[R] EuclideanSpace R (Fin n), (forall x, Q x = x) -> AEMeasurable (grassmannianAction (m := m) Q) /\ Measure.map (grassmannianAction (m := m) Q) = theorem mattila_10_10_plane_sections {n m : N} {t : R} {A : Set (EuclideanSpace R (Fin n))} [MeasurableSpace (Grassmannian n (n - m))] (: Measure (Grassmannian n (n - m))) (h : IsInvariantGrassmannianMeasure) (hmt : (m : R) < t /\ t < n) (hA : MeasurableSet A) (hAfinite : muH[t] A < inf) (hApos : 0 < muH[t] A) : let slice := fun (W : Grassmannian n (n - m)) (a : W.1) => A inter {x x - (a : EuclideanSpace R (Fin n)) in W.1} (forall W : Grassmannian n (n - m), forall^m a d(muH[(m : R)] : Measure W.1), (muH[t - m] : Measure (EuclideanSpace R (Fin n))) (slice W a) < inf) /\ forall^m W : Grassmannian n (n - m) d, 0 < muH[(m : R)] {a : W.1 dimH (slice W a) = ENNReal.ofReal (t - m)} := by sorry </pre>
14.10. Theorem. Let s be a positive number. Suppose that there exists a Radon measure μ on $\mathbb{R}^n$ such that the density $\Theta_s(\mu, a)$ exists and is positive and finite in a set of positive μ measure. Then s is an integer.	Geometric measure theory	<pre> theorem mattila_14_10_marstrand_density_integer {n : N} {s : R} {mu : Measure (EuclideanSpace R (Fin n))} (hs : 0 < s) (hmu : IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu) (hdensity : exists E : Set (EuclideanSpace R (Fin n)), 0 < mu E /\ forall x in E, exists theta : R>=0inf, 0 < theta /\ theta < inf /\ Tendsto (fun r : R => mu (closedBall x r) / ENNReal.ofReal ((2 * r) ^ s)) (nhds[>] 0) (nhds theta)) : exists m : N, s = m := by sorry </pre>
15.3. Definition. A set E subset $\mathbb{R}^n$ is called m-rectifiable if there exist Lipschitz maps $f_i : \mathbb{R}^m \rightarrow \mathbb{R}^n$, $i = 1, 2, \dots$, such that $H^m(E \setminus \bigcup_i f_i(\mathbb{R}^m)) = 0$. 15.7. Definition. We say that a subset E of $\mathbb{R}^n$ is m-linearly approximable if for H^m almost all a in E the following holds: if ϵ is a positive number, there are positive numbers r_0 and λ and an affine m-plane W such that a in W and for any $0 < r < r_0$, $H^m(E \cap B(x, \epsilon r)) \geq \lambda r^m$ for x in $W \cap B(a, r)$, and $H^m(E \cap B(a, r) \setminus W(\epsilon r)) < \epsilon r^m$. 15.17. Definition. Let A subset $\mathbb{R}^n$, a in $\mathbb{R}^n$ and V in $G(n, m)$. We say that V is an approximate tangent m-plane for A at a if $\Theta^m(A, a) > 0$ and for all $0 < s < 1$, $r^s H^m(A \cap B(a, r) \setminus X(a, V, s)) \rightarrow 0$ as $r \rightarrow 0$. 15.19. Theorem. Let E be an H^m measurable subset of $\mathbb{R}^n$ with $H^m(E) < \infty$. Then the following are equivalent:	Geometric measure theory	<pre> def Grassmannian (n m : N) := {V : Submodule R (EuclideanSpace R (Fin n)) // Module.finiteRank R V = m} def RectifiableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop := exists f : N -> EuclideanSpace R (Fin m) -> EuclideanSpace R (Fin n), (forall j, exists K : NNReal, LipschitzWith K (f j)) /\ muH[(m : R)] (E \ union j, range (f j)) = 0 def LinearlyApproximableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop := forall^m a dmuH[(m : R)].restrict E, forall eta : R, 0 < eta -> exists V : Grassmannian n m, exists r0 c : R, 0 < r0 /\ 0 < c /\ forall r : R, 0 < r -> r < r0 -> (forall x, x - a in V.1 -> x in closedBall a r -> </pre>

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(1) E is m -rectifiable. (2) E is m -linearly approximable. (3) For $\mathcal{H}^m$ almost all a in E there is a unique approximate tangent m -plane for E at a . (4) For $\mathcal{H}^m$ almost all a in E there is some approximate tangent m -plane for E at a .

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ENNReal.ofReal (c * r ^ m) <= muH[(m : R)] (E inter closedBall x
(eta * r))) /\
muH[(m : R)] (E inter closedBall a r inter
{x | eta * r <= infDist (x - a) (V.1 : Set (EuclideanSpace R (Fin
n)))) <
ENNReal.ofReal (eta * r ^ m)

def IsApproximateTangentPlane {n m : N} (E : Set (EuclideanSpace R (Fin n)))
(a : EuclideanSpace R (Fin n)) (V : Grassmannian n m) : Prop :=
0 < upperHausdorffDensity (m : R) E a /\
forall s : R, 0 < s -> s < 1 ->
Tendsto (fun r : R =>
muH[(m : R)] (E inter closedBall a r inter
{x | s * dist x a <= infDist (x - a)
(V.1 : Set (EuclideanSpace R (Fin n))))} /
ENNReal.ofReal (r ^ (m : R))) (nhds[>] 0) (nhds 0)

theorem mattila_15_19_rectifiability_tangent_planes
{n m : N} {E : Set (EuclideanSpace R (Fin n))}
(hEmeas : MeasurableSet E) (hEfinite : muH[(m : R)] E < inf) :
List.TFAE [RectifiableSet n m E, LinearlyApproximableSet n m E,
forall^m a dmuH[(m : R)].restrict E,
exists! V : Grassmannian n m, IsApproximateTangentPlane (m := m) E
a V,
forall^m a dmuH[(m : R)].restrict E,
exists V : Grassmannian n m, IsApproximateTangentPlane (m := m) E a
V] := by
sorry
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